

GALLANT IAS ACADEMY

PRELIMS CUM MAINS BATCH 2027

ENVIRONMENT

MODULE 1

BASICS OF ECOSYSTEM

ENVIRONMENT

It is the sum total of all surroundings of a living organism, including physical, biological and chemical factors, which provide conditions for development and growth of that living organism. Environment means all the **living and non-living things around us** such as animals, plants, forests, farms, continents, oceans, clouds, ice caps, rivers, structures, urban centres, living places, etc.

ENVIRONMENTAL SCIENCE

It is the scientific study of the environment, analysing the interaction between chemical, biological, and physical elements of the environment and their impact on ecosystems and the organisms. We need to understand our interaction or relationship with the environment to creatively solve environmental problems. That's why environmental science is of prime importance. Identifying a problem is the first step in solving it, and environmental science does that.

SPECIES

There are different concepts of species: morphological, genetic, and biological.

Definition

A species is a specific group of organisms whose members share many common characteristics, including one or more of the following:

	<u>Criterion</u>
1	Having morphological similarity
2	Genetic similarity
3	Common ancestry
4	Ecological niche similarity (Ecological Niche - Role of a species in an ecosystem)
5	Reproductive isolation from other species (Biological species concept : Developed by T. Dobzhansky; popularised by Ernst Mayr)

ECOLOGICAL HIERARCHY

- **Individual / Organism** - an individual living being that has the ability to act or function independently. It may be a plant, animal, bacterium, fungus, etc.
- **Population** - all organisms of a species at one place; a group of single individuals having the potential to interbreed and produce fertile offspring (e.g., the entire elephant population). It is a group of organisms usually of the same species, occupying a defined area during a specific time.
- **Biological Community - All organisms at a place.** Total of all species/populations

inhabiting a particular area; the total flora and fauna (Biota). Note: multiple communities can exist in a single place.

ECOLOGY

It is an in-depth analysis of the relationships of living organisms with each other and with the environment. Derived from two Greek words, **Oikos** (home or habitation) and **Logos** (study or discourse). Hence, literally speaking, ecology is the study of organisms in their own habitat. Broadly divided into Autecology and Synecology.

Autecology deals with the relationship of one species of organism. **Synecology** deals with the ecological study of communities or an entire ecosystem.

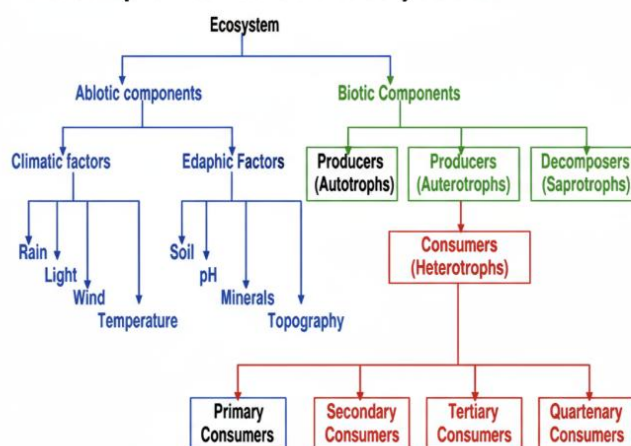
Ecology is the study of the **structure and function of an Ecosystem.**

ECOSYSTEM

The concept of ecosystem was first proposed by **A. G. Tansley (1935)**. The fundamental principle of the ecosystem is that at any place where organisms live, there is a continuous interaction between plants, animals and their environment to produce and exchange materials. The organisms of any community, besides interacting among themselves, always have functional relationships with the external world or the environment. This **“structural and functional system of communities of living organisms and their environment is called an ecological system, or in short, the ecosystem”**.

“The biotic (living) community and its physical (non-living) environment, in which matter cycles and energy flows, are called ecosystems.” Examples: pond, lake, desert, grassland, forest.

Components of Ecosystem



BIOSPHERE

- Sum **total of all Ecosystems.**
- **Thin surface layer on Earth where life exists.**

STRUCTURE OF AN ECOSYSTEM

- **BIOTIC**
- **ABIOTIC**

ECOLOGICAL FUNCTIONS

- **Energy Flow**
- **Nutrient Cycling**
- **Ecological Succession**
- **Feedback Mechanism / Regulation**

PRODUCERS / AUTOTROPHS

Producers are **organisms that can prepare their own food from simple inorganic substances like carbon dioxide and water in the presence of sunlight**. They are self-nourishing. Terrestrial ecosystems have trees, shrubs, herbs, grasses, and mosses that contribute in varying degrees to production. Algae of various types are the most important producers of aquatic ecosystems, although in estuaries and marshes, grasses may also be important producers.

Classified into two types: Phototrophs and Chemotrophs.

<u>Type</u>	<u>How Food is Produced</u>	<u>Examples</u>
<u>Phototrophs</u>	Through photosynthesis, using solar radiation	Green plants, Phytoplankton
<u>Chemotrophs</u>	Through chemosynthesis - oxidation of inorganic substances	Sulphur bacteria, Nitrifying bacteria

CONSUMERS / HETEROTROPHS

They consume the food produced by the autotrophs. Divided into the following types:

<u>Type</u>	<u>Also Called</u>	<u>Food Source</u>	<u>Examples</u>
<u>Herbivores</u>	Primary consumers	Producers: plants and algae	Rabbit, Deer, Elephant
<u>Primary Carnivores</u>	Secondary consumers	Herbivores	Fox, Wolf
<u>Secondary Carnivores</u>	Tertiary consumers	Primary carnivores	Lion, Tiger
<u>Omnivores</u>	-	Both plants and animals	Humans

Phagotrophs are a specific type of heterotroph, not a separate category. **All phagotrophs are heterotrophs, but not all heterotrophs are phagotrophs.** The difference lies in the **method of feeding**: heterotrophs are any organisms that consume other organisms for energy, while phagotrophs are heterotrophs that do so by engulfing or ingesting solid particles or whole organisms.

PHOTOSYNTHESIS

Photosynthesis is one of the most important natural processes; it converts **radiant energy** into chemical energy and links the non-living world with the living world. During photosynthesis, **plants utilise carbon dioxide, water, and sunlight in the presence of chlorophyll to build complex carbohydrate molecules**. In these molecules, **solar energy is trapped as potential chemical energy**. Thus, solar energy is introduced into the biosphere through photosynthesis.

$6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$ (Light energy stored as potential chemical energy)

ECOSYSTEM PRODUCTIVITY

Biomass: Total weight of living matter and organic material per unit area.

Productivity is the rate of biomass production. It is the rate at which chemical energy is produced in an ecosystem, expressed as grams of organic matter per square metre per year. Divided into two types: primary and secondary productivity.

PRIMARY PRODUCTION

It is the **rate of food produced by producers, or the rate of solar energy trapped at the first trophic level.** Expressed in terms of weight or energy. Classified into two - Gross Primary Productivity (GPP) and Net Primary Productivity (NPP).

<u>Type</u>	<u>Definition</u>
<u>Gross Primary Productivity (GPP)</u>	Total amount of energy produced by producers (autotrophs) at trophic level one; i.e., the rate at which an ecosystem's producers convert solar energy into chemical energy as biomass. It depends on the photosynthetic capacity of producers and environmental factors such as temperature, rainfall, and total solar radiation.
<u>Net Primary Production (NPP)</u>	The amount of energy or organic matter fixed or stored at trophic level one (producers) after respiration losses, which is available to higher trophic levels. $NPP = GPP - \text{Respiration losses}$.

There is a **positive correlation between primary productivity and solar radiation.** **Since solar radiation decreases from the equator towards the poles,** primary productivity also, on average (with a few intermediate exceptions), decreases markedly towards the poles.

SECONDARY PRODUCTION

The potential energy from primary production furnishes the energy required by other trophic levels. Some of this energy is consumed by herbivores or omnivores, which may in turn be eaten by carnivores. However, **much of this ingested food is not assimilated; herbivores may assimilate only 10 per cent of the ingested food.** Assimilated energy is used for metabolic processes such as respiration, excretion, and reproduction. The **resultant amount of energy stored in the tissues of heterotrophs is termed net Secondary Production.**

ORDER OF PRODUCTIVITY

<u>Rank</u>	<u>Terrestrial Ecosystem</u>	<u>Aquatic Ecosystem</u>
1 (Most productive)	Swamps, marshes, tropical rain forests	Estuaries
2	Temperate forest	Lakes & streams
3	Northern Coniferous forest (Taiga)	Continental shelf
4	Savanna	Open ocean (least productive)

5	Agricultural land	-
6	Woodland & shrub land	-
7	Temperate Grassland	-
8	Tundra (Arctic & alpine)	-
9	Desert Scrub	-
10 (Least productive)	Extreme Desert	-

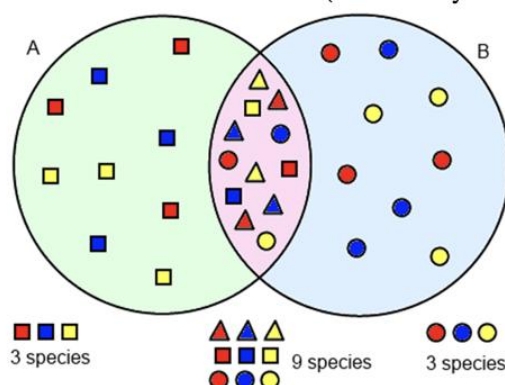
Ecotone

An ecotone is a **zone of junction / confluence or a transition area** between two or more diverse ecosystems.

Examples: **Mangrove forests** (between marine and terrestrial ecosystem); **Grassland** (between forest and desert); **Estuary** (between fresh water and salt water); **Riverbank or marshland** (between dry and wet).

Characteristics of Ecotone

- May be **narrow (e.g., between grassland and forest) or wide (e.g., between forest and desert).**
- Has **intermediate conditions** relative to the adjacent ecosystems; hence it is a **zone of tension.**
- Is a **zone of abundance / higher biodiversity.**
- The number and **population density** of species of an outgoing community generally decreases as one moves away from the community; ecotones are generally more biologically productive areas.
- A well-developed ecotone contains some unique species entirely different from those of the adjoining communities.
- Is a **zone of specialisation.**
- Species in an ecotone have a **broad ecological niche and amplitude.**



Ecocline

- Ecocline is a **zone of gradual but continuous change from one ecosystem to another** when **there is no sharp boundary between the two in terms of species composition.**
- Ecocline occurs across an environmental gradient - **a gradual change in abiotic factors such as altitude, temperature (thermocline), salinity (halocline), depth, etc.**

Edge Effect

- **Edge effect** refers to **changes in population or community** structures that **occur at the boundary of two habitats (ecotone).**
- Sometimes the **number of species and the population density of some species in the ecotone** is much greater than in either adjacent community. This is called the **edge effect.**

ECOTYPE

- An ecotype is a **population of a species that differs genetically from other populations** of the same species because local conditions have selected for certain unique physiological or morphological characteristics.
- **Ecotypes are adapted to survive in specific local habitats.**
- **Ex: Royal Bengal Tiger** - adapted to mangroves; can drink salt water.
- **Examples: Kharai Camel** (Gujarat), Indian Rhino.

Ecad (also called ecophene) refers to a non-genetic, temporary, environmentally-induced variation within a single species, while an **ecotype** refers to a genetically distinct population within a species that has evolved through natural selection to adapt to a specific habitat.

ECOLOGICAL NICHE

[Concept introduced by **Joseph Grinnell**]

Niche refers to the **unique functional role and position of a species in its habitat or ecosystem.**

Ecological niche may also be defined as the **sum total of habitat requirements and behavioural adaptations of a species to persist and produce offspring.**

<u>Type of Niche</u>	<u>Definition</u>
<u>Fundamental Niche</u>	Where an organism could exist without any ecological interactions.
<u>Realised Niche</u>	Where the population actually exists, in the presence of interactions and competition.
<u>Multidimensional Niche</u>	Fundamental niche constrained by limiting factors.

Gause's Principle of Competitive Exclusion

Complete competitors in an ecosystem cannot coexist for long.

HABITAT

- The biotic and abiotic conditions in which a species lives or may live, i.e., the environmental conditions in which a species exists.
- Address of a population.
- The place where an organism lives, including all living and non-living factors or conditions of the surrounding environment.
- Habitat is a population-specific term; however, two or more species/populations may share a common habitat.

ECOLOGICAL AMPLITUDE

The range of environmental conditions (such as temperature, humidity, and light) within which an organism can survive, grow, and reproduce.

BIOTOP

- The address of a community is called its Biotope.

- One ecosystem / Biotop may consist of more than one ecological habitat.

Ecological Guild

A group of related or unrelated species in an ecosystem exploiting resources in a particular manner (e.g., herbivores).

A guild (or ecological guild) is any group of species that exploit the same resources, or that exploit different resources in related ways.

It is not necessary that species within a guild occupy the same, or even similar, ecological niches.

ENERGY FLOW

ENERGY FLOW is the flow of energy from producers to top consumers. **It is unidirectional.** To understand energy flow in an ecosystem, we rely on **trophic level interaction**, which shows how members are connected based on nutritional needs. Energy flows from lower to higher trophic levels, with a loss of energy at each successive trophic level. Trophic level interaction deals with three concepts: **Food Chain, Food Web, and Ecological Pyramids.**

FOOD CHAINS

The biotic factors of an ecosystem are linked together by food. The sequence of who eats whom is called a food chain. **“A food chain is the sequence of who eats whom in a biological community (ecosystem) to obtain nutrition”** - in other words, **the transfer of energy in an ecosystem through a series of organisms by eating and being eaten** constitutes food chains. The position of organisms in the food chain, or **each step of the food chain, is referred to as a Trophic level.**

There are three types of food chain.

A **trophic cascade** is an ecological phenomenon where the removal or introduction of a top predator triggers a chain reaction of changes through the food web.

DETRITUS FOOD CHAIN

This chain starts with dead organic matter and ends with inorganic compounds. Certain organisms depend exclusively on the dead bodies of animals and plants; these **organisms are called detritivores.**

- Leaf litter → Earthworm → Blackbird
- Dead animal → Earthworm → Frog → Snake → Hawk

DECOMPOSITION

Stages

The important steps in decomposition are **fragmentation**, **leaching**, **catabolism**, **humification**, and **mineralisation.**

<u>Stage</u>	<u>Process</u>	<u>Key Outcome</u>
<u>Fragmentation</u>	Detritivores (e.g., earthworm) and scavengers break down organic matter / detritus into smaller particles.	Physical breakdown of detritus.
<u>Leaching</u>	Water carries inorganic nutrients and body fluids down into the soil horizon, where they are precipitated as unavailable salts.	Loss of soluble nutrients into deeper soil layers.
<u>Catabolism</u>	Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. All steps above operate simultaneously on the detritus.	Chemical decomposition.
<u>Humification</u>	Accumulation of a dark brown, colloidal, amorphous substance called humus, which is highly resistant to microbial action and decomposes at an extremely slow rate.	Humus serves as a nutrient reservoir; essential for soil fertility.
<u>Mineralisation</u>	Microbes further degrade humus, releasing inorganic nutrients back into the soil.	Return of nutrients to the soil.

Note:

- In a given climatic condition, **decomposition rate is slower if detritus is rich in lignin and chitin, and quicker if detritus is rich in nitrogen and water-soluble substances like sugars.**
- Temperature and soil moisture are the most important climatic factors regulating decomposition. Warm and moist environments favour decomposition, whereas low temperature and anaerobiosis inhibit it, leading to a build-up of organic materials.

FOOD WEB

Food chains in an ecosystem are **interconnected with each other** to form a network called a **food web**. If the links in the chains that make up the web of life are **disrupted due to human activities that lead to the loss or extinction of species**, the web breaks down.

ECOLOGICAL PYRAMID

- **Diagrammatic expression of Trophic levels.**
- **Graphical presentation of the relationship between various trophic levels based on Energy, Biomass, or Numbers.**

<u>Pyramid</u>	<u>Basis</u>	<u>Shape</u>	<u>Exception / Notes</u>
<u>Pyramid of Energy</u>	Amount of energy at each trophic level	Always upright - energy is lost at each successive trophic level (respiration, metabolism, locomotion).	No exception. Most reliable pyramid.
<u>Pyramid of Biomass</u>	Total mass (dry weight) of living organisms at each trophic level at a given time. Standing Crop / Biomass - total biomass at a specific point in time.	Generally upright.	Inverted in aquatic ecosystems (e.g., pond, lake), where phytoplankton are the main primary producers but have high turnover rates.
<u>Pyramid of Numbers</u>	Total number of individuals of different species at each trophic level.	Upright in grassland and pond ecosystems.	Inverted when parasites are involved (a single tree can host a larger number of parasites than itself). Spindle-shaped when a single large tree supports many herbivores, which are then preyed upon by fewer top predators.

ECOLOGICAL SUCCESSION

The gradual and progressive replacement of one community by another, until a stable community (the climax community) develops in a given area, is called **ecological succession**. The intermediate stages are called **seres**. The final stable community in complete equilibrium with its physical environment is the **climax community**.

<u>Feature</u>	<u>Primary Succession</u>	<u>Secondary Succession</u>
<u>Starting surface</u>	Bare surface with no life or soil (e.g., newly erupted volcanic rock, sand dunes, cliff faces).	Surface where an ecosystem previously existed but has been disturbed or disrupted (e.g., after logging, fire, flood, storm, earthquake, or volcano).
<u>Soil at start</u>	No soil present.	Soil already present.

Pioneer species	Lichens, algae - organisms that do not require soil.	Different pioneer species than primary succession; varies with the type of disturbance.
Rate	Very slow; can take several hundred years to produce fertile soil naturally.	Faster, as soil and residual organic matter are already present.
Example	Bare lava rock → lichens → mosses → grasses → shrubs → trees.	Abandoned farmland → weeds → shrubs → forest.

Pioneer Species

The first organisms to colonise any newly available area and begin the process of ecological succession.

Climax Community

A mature, stable community that is the final stage of ecological succession. The conditions continue to be suitable for all members of the community until another disturbance clears the area. Eg. Pine trees in a forest ecosystem.

LICHENS

Lichens do not require soil. They are composed of two organisms - algae and fungi - in a symbiotic relationship: the algae photosynthesise, while the fungi absorb nutrients from rocks and retain water.

Over time, lichens break down rock; the forces of weathering and erosion reduce it further into smaller pieces. When lichens die, they accumulate in cracks and decompose, adding small amounts of organic matter to form a rudimentary soil. Simple plants such as mosses and ferns can then grow in this new soil. As these simple plants die, they add more organic material, leading to the creation of fertile soil - made up of broken rock, decayed organisms, water, and air. The soil layer thickens; grasses, wildflowers, and other plants begin to take over, followed by shrubs and trees. Insects, small birds, and mammals move in. What was once bare rock now supports a variety of life. Primary succession can take several hundred years to produce fertile soil naturally.

Lichens are good pollution indicators.

ECOLOGICAL INTERACTIONS

<u>Type</u>	<u>Effect</u>	<u>Example</u>
<u>1. Mutualism</u>	Both organisms benefit.	Lichens (algae + fungi)
<u>2. Commensalism</u>	One benefits; the other is unaffected.	Birds nesting in tree hollows
<u>3. Competition</u>	Both organisms are harmed.	Competition between cheetah and lion for food

4. Predation & Parasitism	One benefits; the other is harmed.	Predation: big fish eating small fish. Parasitism: blood-sucking by a tick.
5. Ammensalism	One is harmed; the other is unaffected.	Cattle walking through grass
6. Neutralism	Neither organism benefits nor is harmed.	Cactus and Tarantulas

Previous Year Questions

- Which of the following have species that can establish a symbiotic relationship with other organisms? [2021]
 - Cnidarians
 - Fungi
 - Protozoa
 - 1 and 2 only
 - 2 and 3 only
 - 1 and 3 only
 - 1, 2 and 3
- Which one of the following is the best description of the term 'ecosystem'? [2015]
 - A community of organisms interacting with one another
 - That part of the Earth which is inhabited by living organisms
 - A community of organisms together with the environment in which they live
 - The flora and fauna of a geographical area
- Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms? [2013]
 - Ecotone
 - Ecological niche
 - Habitat
 - Home range
- Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity? [2013]
 - Oceans, lakes, grasslands, mangroves
 - Mangroves, oceans, grasslands, lakes
 - Mangroves, grasslands, lakes, oceans
 - Oceans, mangroves, lakes, grasslands
- With reference to food chains in ecosystems, which of the following kinds of organism are known as decomposer organisms? [2013]
 - Virus
 - Fungi
 - Bacteria
 - 1 only
 - 2 and 3 only
 - 1 and 3 only
 - 1, 2 and 3
- Within biological communities, some species are important in determining the ability of a large number of other species to persist in the community. Such species are called: [2000]
 - Keystone species
 - Allopatric species
 - Sympatric species

d) Threatened species

7. Some species of plants are insectivorous. Why? [2010]
- a) Their growth in shady and dark places does not allow them to undertake sufficient photosynthesis and thus they depend on insects for nutrition.
 - b) They are adapted to grow in nitrogen deficient soils and thus depend on insects for sufficient nitrogenous nutrition.
 - c) They can not synthesize certain vitamins themselves and depend on the insects digested by them.
 - d) They have remained in that particular stage of evolution as living fossils, a link between autotrophs and heterotrophs.
8. A pesticide which is a chlorinated hydrocarbon is sprayed on a food crop. The food chain is: Food crop – Rat – Snake – Hawk. In this food chain, the highest concentration of the pesticide would accumulate in which one of the following? [2010]
- a) Food crop
 - b) Rat
 - c) Snake
 - d) Hawk
9. With reference to food chains in ecosystems, consider the following statements: [2013]
- 1. A food chain illustrates the order in which a chain of organisms feed upon each other.
 - 2. Food chains are found within the population of a species.
 - 3. A food chain illustrates the numbers of each organism which are eaten by others.
- Which of the statements given above is/are correct?
- a) 1 only
 - b) 1 and 2 only
 - c) 1, 2 and 3
 - d) None
10. Which one of the following is the correct sequence of a food chain? [2014]
- a) Diatoms-Crustaceans-Herrings
 - b) Crustaceans-Diatoms-Herrings
 - c) Diatoms-Herrings-Crustaceans
 - d) Crustaceans-Herrings-Diatoms
11. Lichens, which are capable of initiating ecological succession even on a bare rock, are actually a symbiotic association of: [2014]
- a) Algae and bacteria
 - b) Algae and fungi
 - c) Bacteria and fungi
 - d) Fungi and mosses

12. Which of the following are detritivores? [2021]

- 1) Earthworms
- 2) Jellyfish
- 3) Millipedes
- 4) Seahorses
- 5) Woodlice

Select the correct answer using the code given below.

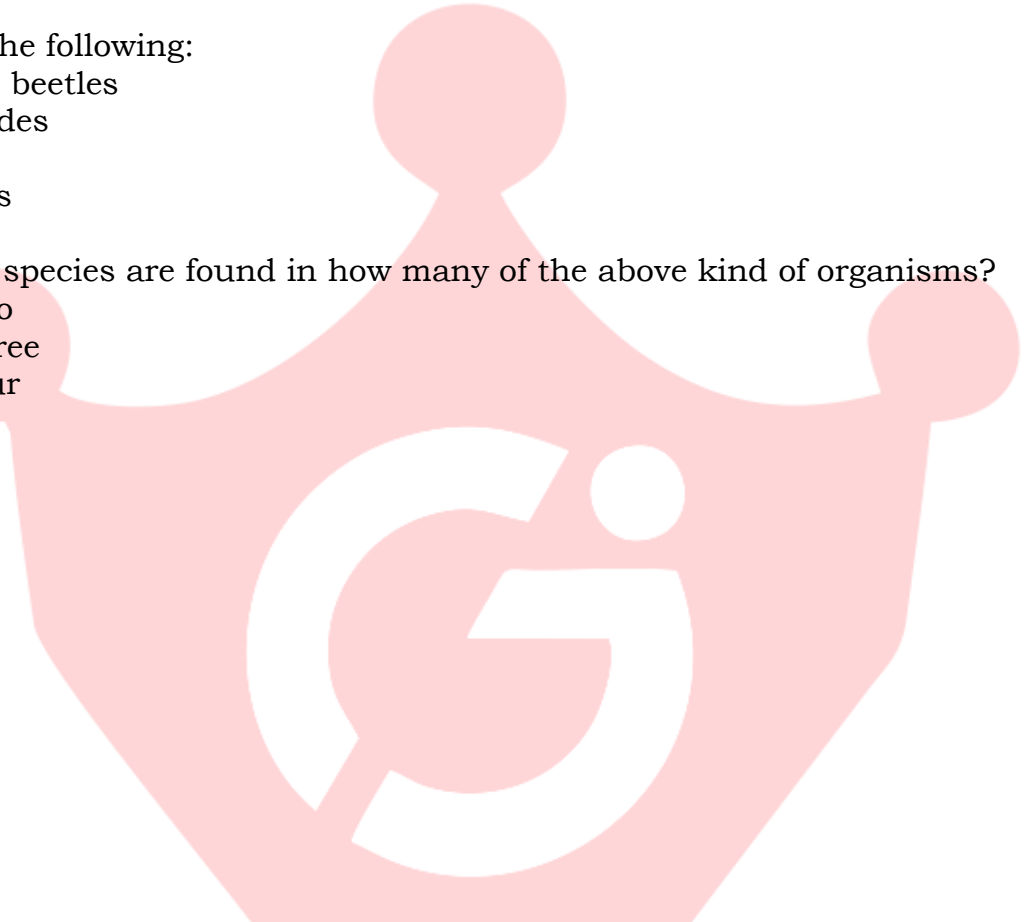
- a) 1, 2 & 4 only
- b) 2, 3, 4 & 5 only
- c) 1, 3 & 5 only

13. Consider the following:

1. Carabid beetles
2. Centipedes
3. Flies
4. Termites
5. Wasps

Parasitoid species are found in how many of the above kind of organisms?

- a) Only two
- b) Only three
- c) Only four
- d) All five



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